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RESEARCH ARTICLE

Crowd Density Estimation Using Enhanced Multi-Column Convolutional Neural Network and Adaptive Collation

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ABSTRACT Accurate crowd density estimation is essential for public safety operations, urban and transportation design and multiple intelligent systems. The research presents an improved Multi-Column Convolutional Neural Network (MC-CNN) structure to predict crowd density. The model uses MobileNet transfer learning capabilities and implements three parallel convolutional columns with different kernel sizes to extract spatial information at different receptive field levels. The design enhances the ability to extract and combine multi-scale features, resulting in better spatial crowd density representation. The model incorporates the output of its columns to improve its understanding of crowd movement patterns. During training, a dynamic collation and zero-padding strategy is introduced to handle variable-sized input images. This approach’s uniform batch shape improves training stability and allows flexible real-world deployment with inconsistent input resolutions. The experimental results show that the proposed architecture performs better, as revealed by its low MAE scores of 12.93 in Part A and 28.56 in Part B of the ShanghaiTech data, outperforming TransCrowd, CSRNet, and SANet. The research provides a functional solution that accurately measures crowd density in different scenarios while overcoming the existing approaches’ restrictions. It fills the academic research gaps in multi-scale feature extraction, efficient data collation and offers new solutions for crowd analysis.

INDEX TERMS Artificial intelligence, crowd detection, convolutional neural network, deep learning, machine learning.

I. INTRODUCTION

Estimating and monitoring crowd density have become paramount in modern society, especially in the case of public events that attract large audiences. As the number of people attending concerts, festivals, and sports games increases, there is an increase in public safety concerns about

overcrowding, emergencies, or security risks. Thus, effective crowd management plans can keep people safe and improve the security mechanisms of the event for participants [1], [2], [3].

Advanced technologies are crucial for detecting and analysing crowd density. Researchers and event managers can better control crowd density using computer vision and machine learning tools, allowing early detection of potential hazards and prevent tragedies. These technologies offer vital

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information to plan for dealing with events that attract many people [4], [5], [6]. Implementing advanced tracking technologies for crowd safety increases public trust and event participation.

The current methods, which use closed-circuit television (CCTV), have detrimental drawbacks in their implementation. Fixed-angle cameras provide restricted observation areas while failing to detect fast-moving crowds or small localised density increases. The power requirements of these systems make them less appropriate for outdoor or temporary installations. The human-based monitoring of CCTV systems leads to delayed responses because it depends on human beings to detect dangerous crowd situations.

The solution to these problems requires implementing computer vision systems as advanced alternatives. Recent developments in AI, including computer vision, provide organizations with practical tools for crowd density management and maintaining public safety during large-scale public events [1], [2], [7], [8]. This study aims to develop a new neural network model that accurately estimates the number of people in a crowd. The objectives of the study are:

1. Optimize convolutional kernel configurations by designing kernels that capture crowd features across multiple scales, enhancing the model's capacity to detect various crowd patterns.
2. Improve feature integration by implementing an optimised feature fusion mechanism to integrate multi-scale features and ensure thorough crowd representation efficiently.
3. Implement sophisticated strategies using complex pooling techniques in each architecture column to augment spatial information retention and enhance model robustness.
4. Introduce a dynamic collation and padding strategy during batch preparation to handle images of variable dimensions, ensuring stable and uniform training by aligning all samples in a batch to the same spatial resolution.
5. Assess the model's effectiveness, performance, and stability via metrics, such as validation mean average error and gradient norm per epoch.

The novelty of this work lies in the proposal of the improved MC-CNN architecture that uses MobileNet for transfer learning and has three parallel convolutional branches to capture multi-scale spatial features. Our approach differs from the existing models in effectively fusing these diverse features to improve the crowd density estimation accuracy. The model outperforms the state-of-the-art methods, including TransCrowd, CSRNet, and SANet, achieving a lower test MAE of 12.93 on Part A and 28.56 on Part B. The model proves its effectiveness for both crowded and empty scenes with minimal computation requirements.

The next sections follow this order. The literature review examines deep learning techniques for crowd counting. The Methods and Materials section explains the ShanghaiTech

dataset and describes the implemented architecture using a lightweight multi-column CNN. The Results section demonstrates the performance comparison between the proposed model and baseline methods through MAE and RMSE evaluation. The Discussion section analyzes the research results while discussing advantages and limitations and suggests potential enhancements for future work. The Conclusion summarizes the research findings while specifying future research avenues.

II. LITERATURE REVIEW

Estimating crowd density is an important computer vision task with public safety, events, and city planning applications. This paper reviews the latest development and methods in crowd density estimation using deep learning and how they have changed over the past decade.

A. EARLY APPROACHES

The first way to estimate a crowd's density was based on feature extraction and regression models. Lempitsky and Zisserman [9] introduced a foundational density map function, which became instrumental in crowd counting later. Chan et al. suggested a way to protect people's privacy that did not involve tracking specific individuals. They did this using holistic traits, which counted the number of people without violating their privacy [10]. Ryan et al. used multiple local features to improve counting accuracy. They combined different sight cues to increase the reliability of the counting process [11], which are more advanced techniques built on top of the basic ones.

B. DEEP LEARNING ERA

The advent of deep learning marked a significant shift in crowd density estimation methodologies. Zhang et al. introduced a deep convolutional neural network (CNN) for cross-scene crowd density estimation, substantially improving accuracy by leveraging deep learning to generalize across different scenes [3], [12], [13]. This approach was further refined by Boominathan et al. with Crowdnet, which utilized deep CNNs designed explicitly for dense crowd density estimation, achieving higher accuracy in highly congested scenes [5].

C. ADVANCED CNN ARCHITECTURES

Later studies focused on improving CNN architecture to handle the complicated task of estimating crowd density in a better way. The pioneering work of Zhang et al. [14] introduced the first multi-column CNN architecture for crowd counting, establishing the basis for scale-aware designs. Sam et al. created a CNN that switched between networks automatically to find the best one for each crowd density, improving accuracy in all situations [4], [15]. Zhang et al. suggested a scale-adaptive CNN to deal with the problem of crowd images of different sizes. This would let the network respond to different densities and points of view [6].

Cao et al. created a scale aggregation network to improve the accuracy and efficiency of estimating crowd density [16]. This network combined features at different scales.

D. ATTENTION MECHANISMS AND SEGMENTATION

Recent advancements incorporated attention mechanisms and segmentation to enhance crowd density estimation accuracy. Wang and Breckon utilised segmentation-guided attention networks to focus on relevant regions in the image, coupled with curriculum loss to improve the learning process. This approach marked a significant step towards more precise crowd density estimation by ensuring that the network paid attention to the most informative parts of the image [17].

Furthermore, RCCFormer, a Transformer-based architecture proposed by Chen et al. [19], has achieved state-of-the-art crowd counting tasks by effectively capturing long-range dependencies and spatial relationships in dense scenes. Its robust design addresses complex occlusions and density variations more accurately than CNN-based models.

In addition to vision-only transformer models, multimodal approaches also emerged. TransFusion, proposed by Li et al. [20], introduced a transformer-based multimodal fusion framework that combines visual signals with wireless information (e.g., WiFi), enhancing the accuracy and robustness of crowd counting in complex environments.

Another notable transformer-based solution is TAPNet by Zhang et al. [21], which applies dual-optical attention fusion to improve crowd head point counting and localization. TAPNet demonstrates strong performance in highly congested scenes by leveraging fine-grained visual attention mechanisms, effectively detecting and localizing individuals in dense crowds.

E. REAL-TIME AND CROSS-SCENE APPLICATIONS

Real-time crowd density estimation has also seen significant progress. Gündüz and Işık proposed a YOLO-based method for real-time crowd detection, highlighting the performance of various YOLO models in achieving high accuracy and speed [8]. Ryu and Song introduced a method combining crowd density estimation with individual localisation using pseudo-square labels, allowing for personal and localisation in real-time scenarios [22].

Multiple detailed assessments of crowd density estimation technology exist alongside descriptions of its future development. Li et al. conducted an in-depth evaluation of crowd density estimation approaches, including their applications and operational principles (ref1). Khan et al. examined the present trends and projected paths for crowd density measurement methods. They also pointed out. The problems that require solutions include the complexity of various crowd scenes, the variability in crowd densities, the limitations of existing models under different environmental conditions, and the requirement for improved evaluation metrics and benchmarks to assess model performance thoroughly (ref2). In 2022, Hassen et al. examined convolutional neural networks

and heuristic methods in detail (ref10) and presented the advantages and disadvantages of each technique. The heuristic methods demonstrate several benefits, especially when working with sparse data and requiring minimal resources, making them suitable for restricted systems. The methods demonstrate weak points in their ability to handle both dense and complex crowd scenarios. The ability of CNN-based models to handle dense and congested crowds stemmed from their robustness and powerful feature extraction capabilities. These models need extensive training data, perspective maps and computing power, complicating their implementation process. Liu et al. and Patel and Kumar wrote evaluative assessments about learning models and intelligent methods for crowd density estimation, which included convolutional neural networks (CNNs), recurrent neural networks (RNNs) transformer-based architectures, hybrid models, density map regression, object detection and feature extraction techniques (ref12,ref13).

The research has recently developed innovative solutions to advance crowd density estimation methods. Babu Sam et al. created the dilated convolutional neural network CSRNet for processing congested scenes which led to enhanced density estimation accuracy in crowded areas (ref19). The MC-CNN developed by Zhang et al. implemented multiple columns with different receptive fields to manage crowds at various density levels (ref19). The breakthroughs enabled developers to create more flexible and durable systems for crowd density estimation. In their research from 2022 Ali Hassen et al. investigated the integration of CNNs with heuristic methods and presented a systematic evaluation of these combined approaches (ref10). Şirin Gündüz and Işık conducted an analysis of YOLO models' performance for real-time crowd detection while emphasizing the necessity of speed and accuracy for practical use cases (ref11).

Table 1 shows evaluation results for multiple deep learning-based crowd estimation models tested on the ShanghaiTech dataset. The Switching CNN and Scale-Adaptive CNN made the initial attempts to address scale variation and crowd density diversity. The SANet model performed exceptionally well in both Part A and Part B sections by successfully combining features at various scales. YOLO-based models prioritize real-time processing but their performance evaluation on the ShanghaiTech benchmark remains restricted. Our proposed architecture uses scale-sensitive CNNs and detection-based systems to motivate its development because it provides efficient adaptability with high accuracy for various crowd densities.

The advancement of crowd density estimation through deep learning approaches mainly relies on convolutional neural networks (CNNs) to recognize spatial relationships between features that relate to crowd dynamics. The high performance of CSRNet and SANet architectures emerges from their deployment of dilated convolutions together with multi-scale feature aggregation. The models encounter challenges when dealing with shifting crowd densities and occlusions in processing real-world dynamic scenes.

Multi-column CNN approaches perform poorly when they process extracted information during transitions between extreme density variations and complex spatial configurations.

The large computational needs of these models function as obstacles to their implementation in real-time applications and limited resource systems. The current models demonstrate a poor ability to transfer knowledge between different crowd environments during the shift from sparse to dense population conditions. These constraints reveal an urgent requirement to build architectures that will unite high accuracy performance with generalization abilities and fast computation.

The architecture integrates MobileNetV2 as a lightweight encoder for feature extraction with a multi-column CNN density estimation head. The proposed architecture overcomes previous model deficits through its ability to perform well across different crowd densities and real-time processing capabilities. The proposed method achieves comparable performance to state-of-the-art models during evaluation on the ShanghaiTech dataset, especially when evaluating dense crowds.

III. METHODS AND MATERIALS

The following section explains the methodological framework which uses a convolutional neural network architecture to perform crowd density estimation. The ShanghaiTech dataset serves as our training and evaluation dataset because it contains densely populated crowd scenes, enabling the model to learn robust generalization across different environments. The proposed Multi-Column Convolutional Neural Network (MC-CNN) architecture uses MobileNet as its backbone to achieve efficient feature extraction while describing its architecture. The model uses three parallel convolutional columns to extract spatial information at different scales before combining these features through a 1×1 convolution to generate a unified density map. The loss function definition in this section explains the training optimization process, which relies on Mean Squared Error (MSE) to measure the difference between predicted and actual density distributions. The system consists of multiple components that work together to provide an end-to-end solution for precise and efficient crowd counting in real-world complex environments.

A. DATASET

The ShanghaiTech dataset [3], comprising dense (Part A) and sparse (Part B) crowd scenes, was used to train and evaluate the model. This diversity ensures better generalization across varying crowd densities. Data augmentation techniques, such as horizontal flipping and brightness adjustments, were applied to enhance robustness.

The ShanghaiTech dataset is a prominent benchmark for estimating and studying crowd density. The composition comprises Part A and Part B, each featuring a range of crowd scenes in diverse urban environments. This research



FIGURE 1. Sparsely populated crowd scene.

utilized both Part A and Part B datasets to train and evaluate the model. Part A comprises 482 images featuring 241,677 annotated individuals, showcasing denser crowds, making the model better at dealing with dense crowds and poorer in sparsely populated situations. Training on denser crowds also requires more computing power because such a task is more complicated.

Part B contains 716 images featuring 88,488 annotated individuals, representing sparse to moderately dense scenes. Using dataset allowed the model to generalise effectively across a wide range of crowd densities—from sparse public squares and parks (Part B) to extremely crowded urban areas (Part A). While training on dense crowds (Part A) demands more computational power due to its complexity, its inclusion enhanced the model's ability to handle challenging, real-world dense crowd scenarios. Part B, on the other hand, contributed to robust performance in lower-density environments and provided broader contextual diversity. The images include urban settings such as streets, parks, and public squares, enhancing model generalisation in various environments. Diversity facilitates a model trained to manage various crowd density levels effectively. The dataset consists of ground truth density maps generated through Gaussian kernel density estimation, which helps train and evaluate the predictive models. It is more suitable for model training that requires generalisation across diverse crowd conditions, and combining both parts is critical for real-world applications characterised by huge variations in crowd densities. Thus, this study adopts Part B data. This study used data augmentation to increase the number of data samples, specifically implementing horizontal flips with a 50% probability, adding random brightness and contrast, and applying Gaussian blur.

Figures 1 and 2 show examples from the collection. Compared to Figure 1, Figure 2 shows a denser crowd and a higher chance of accidents. It includes annotated data illustrating everyone in the photograph, as shown in Figure 3, the annotated version of Figure 1. Figure 4 displays a density map derived from Figure 1, which adopts a Gaussian filter to portray each individual as a point on the map.

TABLE 1. Comparison of deep learning methods on ShanghaiTech dataset.

Citation	Method	Year	Dataset	MAE	MSE	Key Features
[4] D. B. Sam et al. (2017)	Switching CNN	2017	UCF_CC_50, ShanghaiTech	Part A - 90.4; Part B - 21.6	Part A - 135.0; Part B - 33.4	Switching CNN for crowd counting, handles varying crowd densities
[6] L. Zhang et al. (2018)	Scale-Adaptive CNN	2018	ShanghaiTech, UCF_CC_50	Part A - 86.8; Part B - 16.2	Part A - 139.2; Part B - 25.8	Scale-adaptive convolution for more accurate crowd density estimation
[16] X. Cao et al. (2018)	SANet	2018	ShanghaiTech, UCF_CC_50, UCSD, WorldExpo'10	Part A - 67.0; Part B - 8.4	Part A - 104.5; Part B - 13.6	Scale aggregation network for accurate crowd counting
[8] M. Şirin Gündüz et al. (2023)	YOLO-based Detection	2023	MS COCO	-	-	Real-time crowd detection using YOLO-based models

**FIGURE 2.** Densely populated crowd scene.

B. COLLATION

The custom mini-batch collation strategy we introduce enhances model training stability and minimizes Mean Absolute Error (MAE). The dataset contains images with different resolutions, so naive resizing or cropping methods could damage spatial features and make density annotations unreliable. The custom collate function uses dynamic padding to maintain spatial consistency and density integrity throughout the entire batch.

Let $\mathcal{B} = \{(X^{(j)}, D^{(j)}, C^{(j)})\}_{j=1}^B$ denote a mini-batch of size B , where $X^{(j)}$ is the image tensor, $D^{(j)}$ is the corresponding density map, and $C^{(j)}$ is the ground truth count for the j -th sample. We define the maximal height and width in the batch as:

$$H_{\max} = \max_j H^{(j)}, \quad W_{\max} = \max_j W^{(j)} \quad (1)$$

Each image and its associated density map are padded to these dimensions:

$$\tilde{X}^{(j)} = \text{Pad}(X^{(j)}, H_{\max}, W_{\max}) \quad (2)$$

$$\tilde{D}^{(j)} = \text{Pad}(D^{(j)}, H_{\max}, W_{\max}) \quad (3)$$

where $\text{Pad}(\cdot)$ represents zero-padding applied to the right and bottom borders, preserving semantic meaning of the crowd regions. The padding is applied such that:

$$\tilde{D}^{(j)}(u, v) = \begin{cases} D^{(j)}(u, v), & \text{if } u \leq H^{(j)}, v \leq W^{(j)} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

This padding strategy ensures that the density values outside the original image region remain zero, preserving the correctness of the total count:

$$\sum_{u,v} \tilde{D}^{(j)}(u, v) = C^{(j)} \quad (5)$$

Equation (5) ensures that crowd count supervision remains valid after padding, thereby preventing distortion of learning signals during training. This approach leads to a more stable optimization trajectory and significantly improves MAE, especially when working with images of varying resolutions and aspect ratios. The batched padded tensors are then stacked to form the final mini-batch for model input. This custom batching strategy is essential to maintain spatial and density consistency, and plays a vital role in enhancing model performance in diverse and complex real-world scenarios.

C. MODEL ARCHITECTURE

Figure 3 shows the improved Multi-Column Convolutional Neural Network (MC-CNN) for estimating crowd density's architecture. The network includes three parallel convolutional columns that capture features at various scales. Before going through the set of columns, transfer learning is used, specifically the lightweight MobileNet architecture [26], to extract features from the input image. Ours use this backbone, combining the depth benefits shown by Simonyan and Zisserman [27] with efficient convolution operations, preserving feature extraction quality while reducing computational overhead for real-time applications. MobileNet is based on CNN, which effectively captures many features while being more efficient for real-time applications [12]. Afterwards, the data of extracted features is passed to the

set of columns. Using three parallel columns in the network for crowd density estimation is a strategic design choice that significantly enhances the model's ability to capture a wide range of features [18].

To provide visual clarity on the model's input supervision and output predictions, Figures 4 and 5 illustrate representative examples of annotated inputs and their corresponding predicted density maps.

Figure 4 shows crowd images with annotated ground truth points, where red dots indicate individual head positions used for model training and evaluation. These annotations serve as the basis for generating the ground truth density maps via Gaussian kernels.

Figure 5 presents the proposed MC-CNN model's predicted density maps. It depicts the crowd density's spatial distribution, brighter regions highlight densely populated areas. The close match between the predicted (DM) and the actual ground truth counts (GT) proves the model's effectiveness in estimating in sparse and dense crowds.

The visualization results show that the MC-CNN performs well under different crowd densities and can accurately learn the people's spatial distribution in actual scenarios.

Each column successfully extracts the features at various spatial scales.

The design of each column enables the extraction of features across multiple spatial scales.

1. The first column detects broad spatial features through the combination of 7×7 and 5×5 kernels followed by a 3×3 convolution. The selected kernel sizes in this configuration enable the detection of low-density and large-scale patterns because they provide broad context modeling capabilities according to [28].
2. The second column uses 5×5 and 3×3 kernels followed by a 3×3 convolution to detect mid-level features. The combination of spatial coverage and detail makes this design suitable for detecting medium-density regions and moderately sized patterns according to [29].
3. The third column applies 3×3 and 3×3 kernels before performing another 3×3 convolution to focus on detailed feature extraction. This configuration detects detailed patterns and dense areas by providing precise location capabilities according to [30].

The outputs from these columns, which encompass diverse spatial information at varying scales, are combined to form a 48-channel feature map. Subsequently, the multi-scale feature map is processed through a fusion layer that adopts a 1×1 convolutional layer to integrate the various feature representations. The fusion process consolidates the multi-channel output to a single-channel crowd density map, effectively representing the overall crowd density in the image. The fusion layer is vital for integrating the information derived from the three columns into a unified representation. It combines features across multiple scales, successfully allowing the model to balance large and fine-grained crowd

structures. While Liu et al. [31] leverage semi-supervised learning to mitigate data scarcity, our approach addresses feature diversity through parallel multi-column convolutions. Their findings on the importance of 'strong models' support our design of lightweight MobileNet backbones combined with multi-scale kernels to balance efficiency and accuracy. This architecture enables the MC-CNN to manage various crowd scenarios successfully and improve its performance in crowd density estimation from sparse to dense crowds.

Each column C_i in the network applies a convolution operation on the input image X .

1. The convolution of each layer l in column i is represented as:

$$F_i^l = \sigma(W_i^l * F_i^{l-1} + b_i^l) \quad (6)$$

where:

- F_i^l is the feature map at layer l in column i ,
- W_i^l represents the convolutional kernel weights of layer l in column i ,
- b_i^l is the bias term for layer l ,
- $*$ denotes the convolution operator,
- σ is the activation function,
- $F_i^0 = X$ is the input feature map.

2. After convolution, the output feature maps from each column i are concatenated along the channel dimension:

$$F_{\text{concat}} = \text{Concat}(F_1^L, F_2^L, F_3^L) \quad (7)$$

where:

- F_i^L is the last-layer feature map in column i ,
- $\text{Concat}(\cdot)$ is the channel-wise concatenation operator,
- L is the number of layers in each column.

The resulting concatenated feature map F_{concat} has 54 channels.

3. The concatenated feature map is passed through a 1×1 convolutional layer to reduce it to a single-channel density map D :

$$D = W_{\text{fusion}} * F_{\text{concat}} + b_{\text{fusion}} \quad (8)$$

where:

- W_{fusion} is the 1×1 convolution weight matrix used to fuse the multi-scale feature maps,
- b_{fusion} is the bias term associated with the fusion layer.

Choosing a 1×1 convolutional fusion layer effectively integrates multi-scale spatial information while maintaining computational efficiency. This strategy reduces the dimensionality of the feature maps and combines different spatial scales into a unified representation, which is crucial for accurate crowd density estimation.

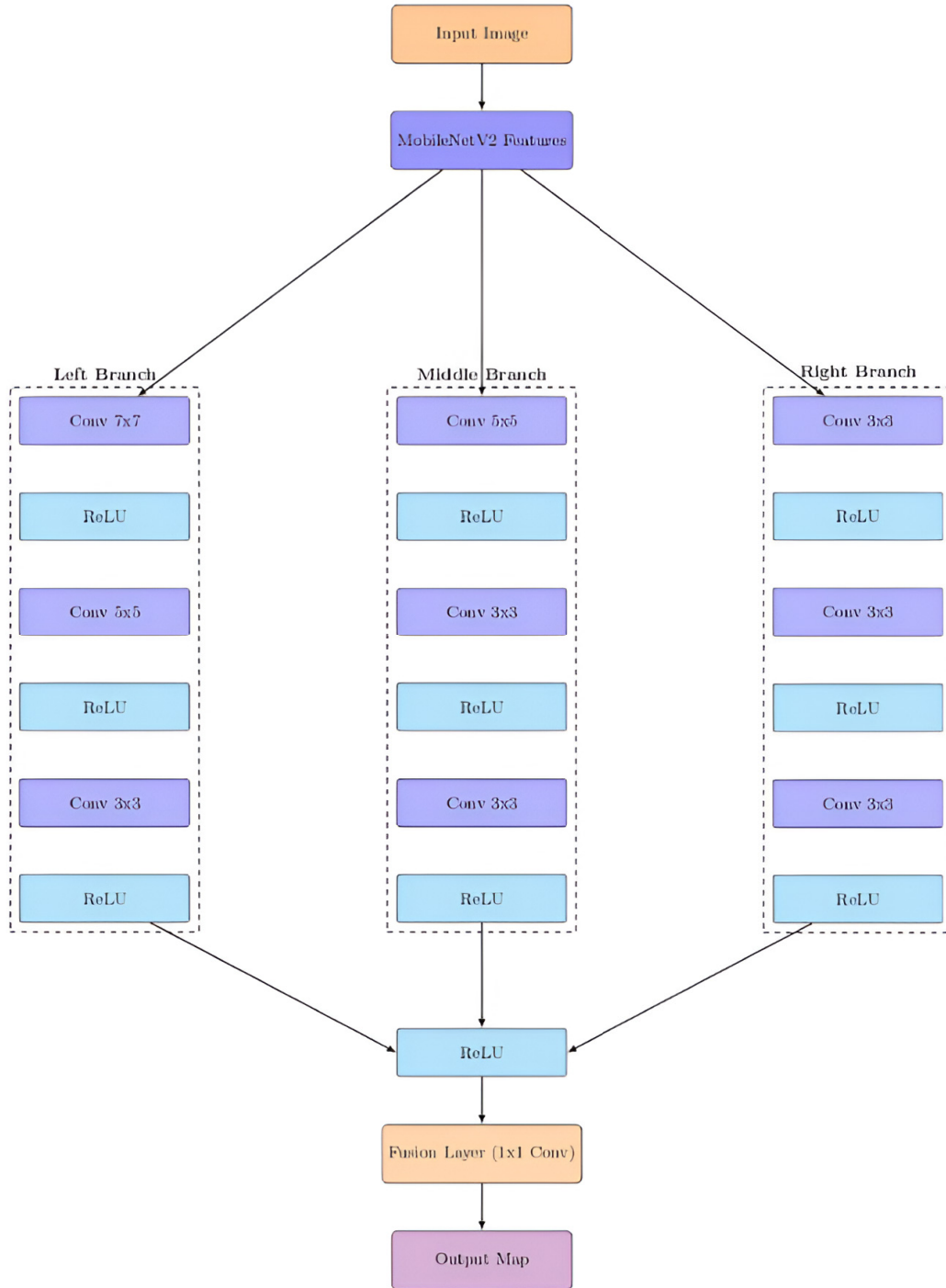


FIGURE 3. Architecture of the proposed enhanced multi-column convolutional neural network (MC-CNN) model, outlining the structure and key components for crowd density estimation.

D. TRAINING

The Mean Squared Error (MSE) loss function was the optimization method to enhance the model's crowd density estimation accuracy. The function penalizes the differences between predicted density maps and actual ground truth values, enabling the model to learn precise image-to-density

value mappings. The loss is defined as:

$$L = \frac{1}{N} \sum_{i=1}^N (D_i - D_{gt,i})^2 \quad (9)$$

where:

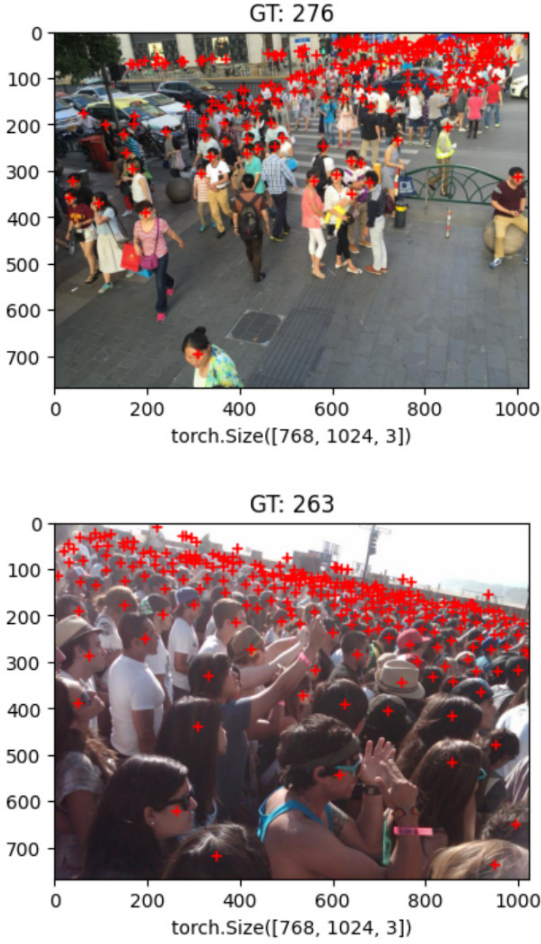


FIGURE 4. Annotated versions of Figure 1 and Figure 2, showing key features and bounding boxes used for crowd annotation and estimation.

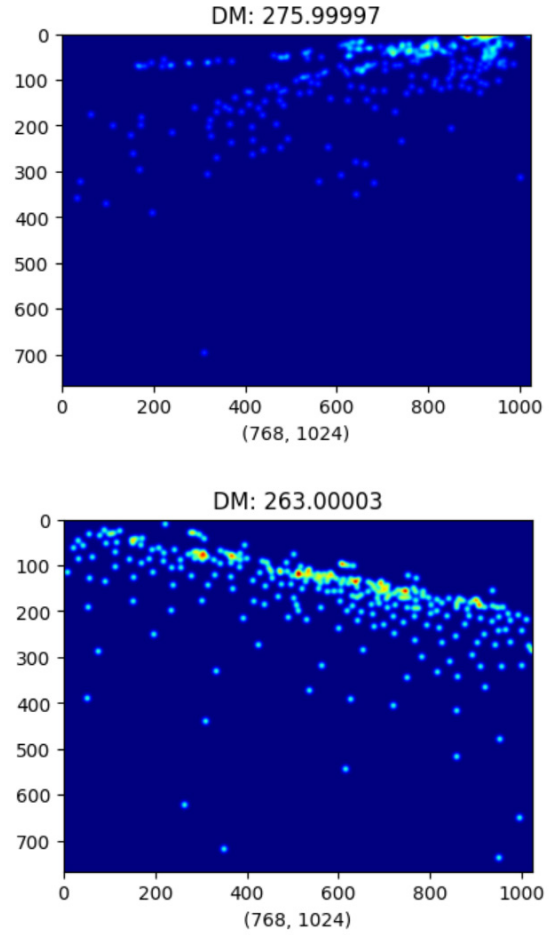


FIGURE 5. Density maps corresponding to Figure 1 and Figure 2, depicting spatial distribution of crowd density in the images. Darker regions indicate zones with higher crowd concentration.

- N is the total number of pixels in the density map,
- D_i is the predicted density value at pixel i ,
- $D_{gt,i}$ is the ground truth density value at pixel i .

The MSE loss (1) effectively penalizes large errors, making it particularly suitable for dense crowd regions where small misestimations may result in significant total count errors. However, to further enhance the model's robustness and account for global counting accuracy, we introduced a combined loss function, which incorporates both pixel-level MSE and count-level Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{combined}} = \lambda \cdot \mathcal{L}_{\text{MSE}}(\hat{D}, D) + (1 - \lambda) \cdot \mathcal{L}_{\text{MAE}}(\hat{C}, C) \quad (10)$$

Here, the predicted count $\hat{C}$ is computed by summing all pixels in the predicted density map:

$$\hat{C} = \sum_{u=1}^H \sum_{v=1}^W \hat{D}(u, v) \quad (11)$$

The hyperparameter $\lambda \in [0, 1]$ is set empirically to 0.8 to balance spatial detail preservation and global count accuracy.

To stabilize training and track convergence, we additionally monitored:

- The ℓ_2 -norm of the gradients:

$$\|\nabla \mathcal{L}\|_2 = \left(\sum_j \|\nabla_{\theta_j} \mathcal{L}\|_2^2 \right)^{1/2} \quad (12)$$

- The ℓ_2 -norm of the model weights:

$$\|\theta\|_2 = \left(\sum_j \|\theta_j\|_2^2 \right)^{1/2} \quad (13)$$

Training was conducted using the Adam optimizer with a learning rate of:

$$\eta = 10^{-3} \quad (14)$$

We employed a StepLR scheduler to decay the learning rate by a factor of 0.1 every 30 epochs:

$$\eta_t = \eta_0 \cdot \gamma^{\lfloor \frac{t}{T} \rfloor}, \quad \gamma = 0.1, \quad T = 30 \quad (15)$$

The training process ran for five epochs to prevent overfitting through validation loss monitoring for early stopping.

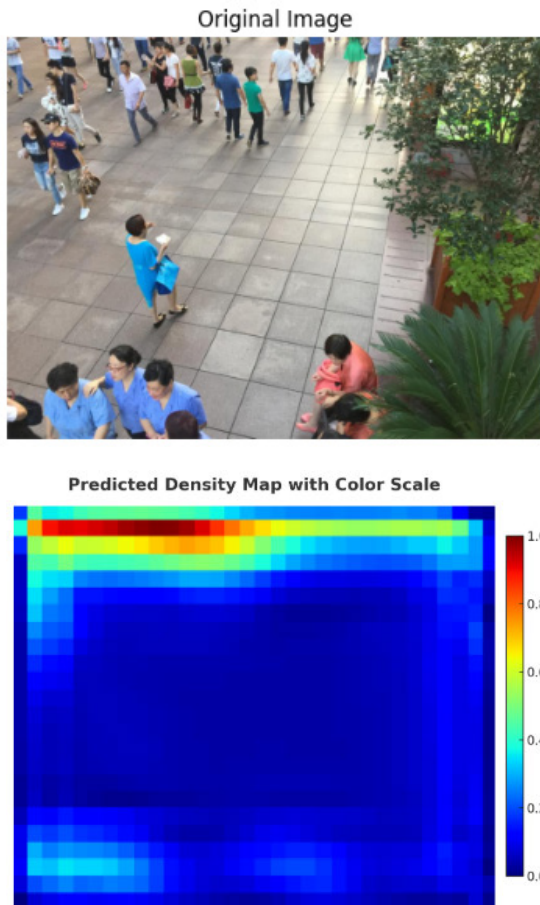


FIGURE 6. Results obtained by applying the proposed model to the test dataset, demonstrating its performance in estimating crowd density.

This research method brings architectural efficiency together with adaptive collation and a dual-focus loss function to realize robust and scalable crowd density estimation. The MobileNet-based multi-column architecture provides a detailed multi-scale representation, while the dynamic batch collation strategy enables efficient input processing. The proposed framework overcomes actual crowd counting challenges effectively by optimizing training and monitoring.

IV. RESULTS

The proposed MC-CNN model produces the results shown in Figure 6 when we used the ShanghaiTech test dataset for estimating crowd density. The first image shows the original crowd scene, and the predicted density map in the following image. The model reaches its best MAE at 17.46 for Part A, proving its capability to estimate dense crowd situations. For Part B, the model achieved a significantly lower best validation MAE of 1.43, confirming its strength in sparse and moderately dense scenes. These results proof the model's effectiveness in estimating various crowd density.

Figure 7 shows that the suggested MC-CNN architecture for crowd density estimation's training mean absolute error

(MAE) decreases progressively, meaning that the model improves during training. The validation MAE, on the other hand, changes, indicating that the model works with unseen data. The difference between the training and validation MAE shows the model's generalizability. If the validation MAE starts to settle or rise while the training MAE continues to decrease, this may indicate overfitting.

The training and validation MAE trends for Part A and Part B of the ShanghaiTech dataset are presented in Figure 7. The evaluation of model performance under different crowd densities becomes possible through combining both parts because Part B contains sparse to moderately dense scenes and Part A contains highly congested scenes. The validation MAE shows fluctuations in Part A because dense crowd estimation faces challenges from occlusions and scale variations. In contrast, Part B shows a consistently decreasing trend in validation MAE, indicating stable generalisation in simpler scenes. The performance stability across both dataset parts can be improved through increased dataset diversity and advanced regularization techniques [23].

The best validation MAE for Part A is 10.70 at epoch 3, while for Part B, it is 1.43 at epoch 4. In both cases, the proposed model outperforms baseline models such as TransCrowd, SANet, and CSRNet. Figure 8 shows the gradient norm of the neural network model during training, which reflects how the weights update across epochs. The X-axis represents the number of epochs, while the Y-axis shows the corresponding gradient norm values.

At the first epoch, the average gradient is very high, reflecting significant updates due to initial random weight adjustment. In the second epoch, the gradient norm drops significantly, indicating a stabilisation of learning, and it remains relatively consistent across the following epochs, with only slight variations. This behaviour is consistent with typical optimisation dynamics, where steep gradients occur early in training, followed by convergence toward more stable updates. This trend demonstrates how the model effectively adapts to the data in sparse and dense crowd conditions.

Figure 9 showcases a neural network's performance in predicting the number of people in the crowd over time. The plot evidenced that it only took seconds for each epoch more than four epochs.

The X-axis displays epoch numbers, and the Y-axis is time duration in seconds. The data points change and show that the processing time for each epoch is different. The third epoch required the shortest time, while the others experienced longer times.

The weight norm evolution during neural network training for crowd density estimation is shown in Figure 10. The weight norm shows minimal change during the first epoch. The weight norm gradually increases during training, indicating the network develops complexity to detect detailed features in crowd images. The weight norm reaches a

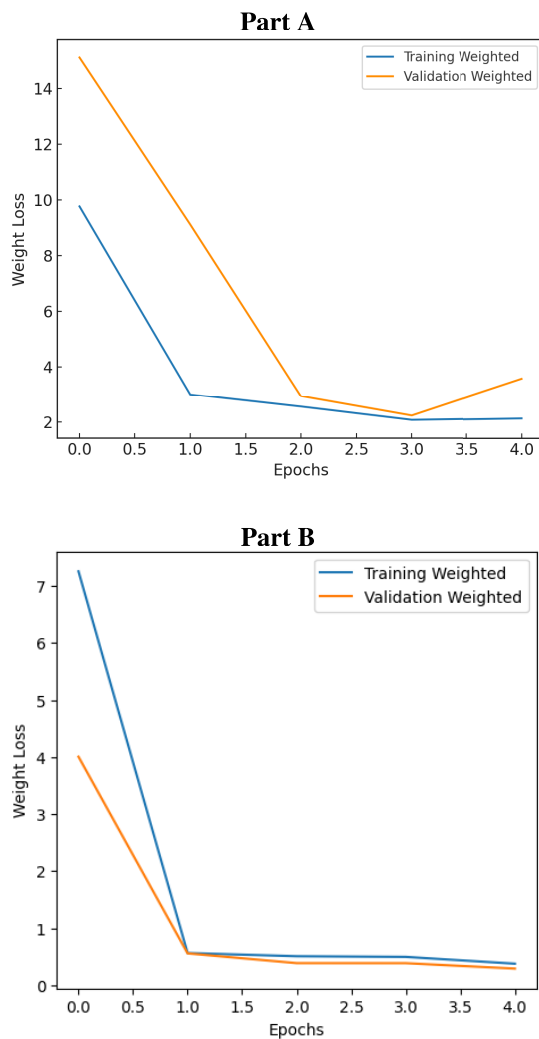


FIGURE 7. Training and validation Mean Absolute Error (MAE) over the epochs, showing the model's learning curve and generalization performance.

stable point or maintains a small variation range, indicating convergence. Li et al. [1] explain that weight stabilization throughout training indicates model convergence because the network learns important features without overfitting excessively. The observed weight stability matches previous studies, which emphasize stable weight dynamics for better generalization performance. Ben Ali Hassen et al. [10] found that weight magnitudes tend to rise progressively because the network detects more complex patterns in input data including crowd density variations.

The distribution of absolute prediction errors for Part A and Part B of the ShanghaiTech test dataset appears in Figure 11. The right side of Part B shows that most prediction errors are concentrated in a low range because the model performs well in sparse and moderately dense scenes. The left side of the Part A distribution shows a broader range of errors because the model faces difficulties with dense and complex crowd scenes that include severe occlusions and scale variations. The red dashed vertical line in both histograms indicates

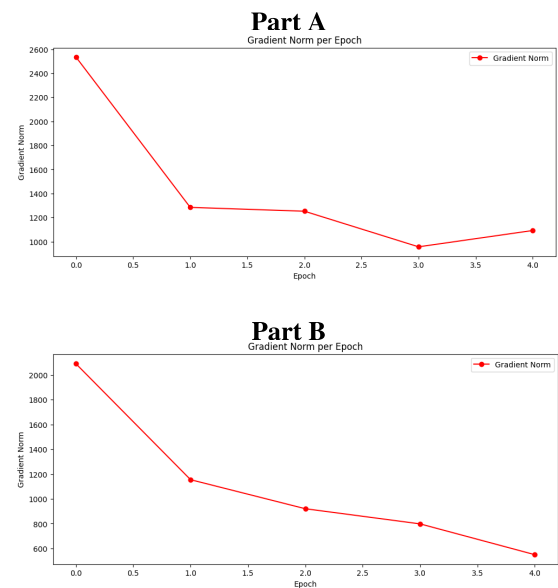


FIGURE 8. Gradient norm in training epochs, providing insights into the model's optimization process stability.

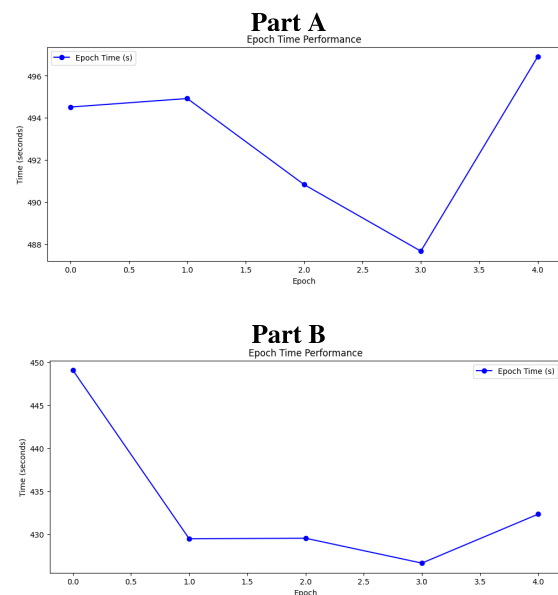


FIGURE 9. Epoch time performance illustrates the model's computational efficiency in the training process.

the mean absolute error (MAE), representing the average prediction error for each validation sample. The purple dotted line indicates the 95th percentile of absolute errors. Most predictions in Part B remain under this threshold, yet Part A occasionally shows significant errors when dealing with extremely dense situations. The Part A MAE score of 10.7 demonstrates strong performance against dense crowd challenges, while the Part B MAE score of 1.43 shows the model excels in less dense environments. The analysis reflects the model's adaptability while showing that dense crowd estimation remains an area for potential improvement.

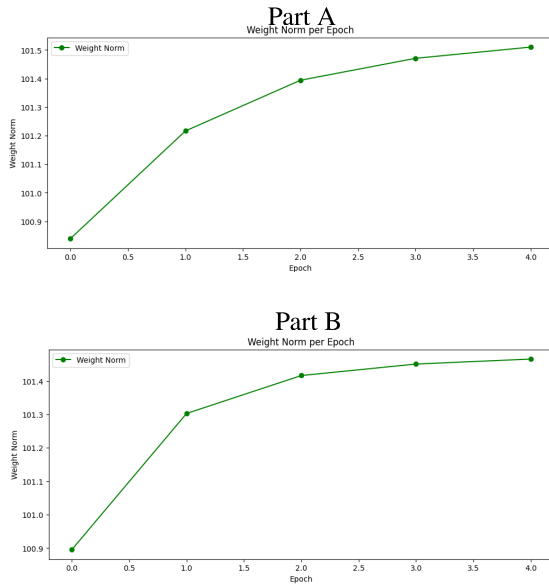


FIGURE 10. Weight norm per epoch, visualising the magnitude of the network weights and providing an indication of model regularisation.

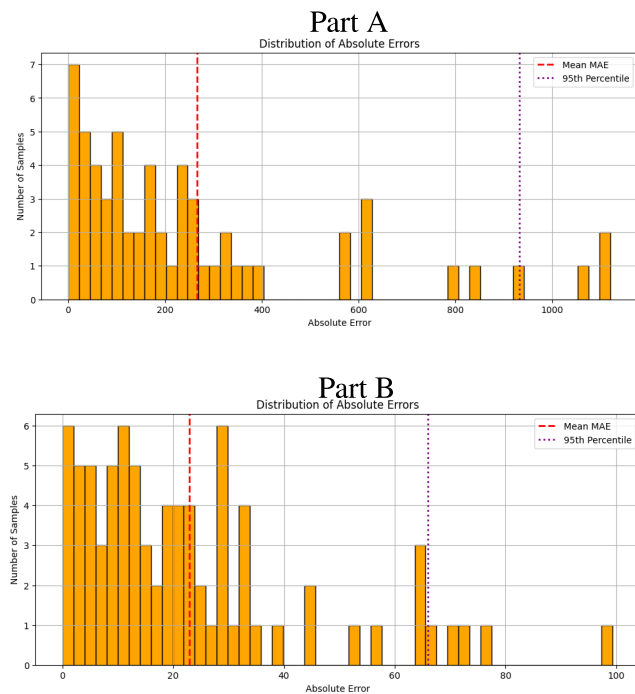


FIGURE 11. Distribution of prediction errors against actual crowd counts.

The discrepancy between a relatively low MAE of 10.7 and a much higher RMSE of 29.88 indicates that while most predictions are close to the ground truth, a small number of large errors (outliers) significantly inflate the RMSE. This is because RMSE disproportionately penalizes larger errors due to the squaring of residuals.

Figure 12 presents several representative cases of such outliers where the model's predictions deviate substantially from the ground truth. These examples help us understand

the nature and sources of high-error cases. In Sample 1, for instance, the ground truth (GT) is 414, but the predicted value is 179.35 only, resulting in an error of over 234. This image contains a highly dense crowd distributed uniformly across the frame. The underestimation might be due to occlusions, high crowd density, or insufficient training examples in similar crowd levels, causing the model to generalize poorly in extreme cases.

Samples 2 and 3 display moderate to high-density scenes, and the model underestimates the count by over 90 in both cases. This again suggests challenges in dense crowd interpretation, possibly due to overlapping individuals and variations in scale or perspective distortion that confuse the network. Sample 4 reveals a negative predicted count (-38.51), a model artifact or a post-processing error (e.g., a regression head with no activation function like ReLU). Such predictions are not only unrealistic but also heavily affect RMSE due to their extreme nature, despite not being frequent. Although Sample 5 shows a medium-sized crowd, the prediction doubles the actual count (163 vs. 79). This may be due to misinterpretation of background textures or a high number of people partially occluded or in the shadows, leading to double counting or misclassification. These samples collectively confirm that the high RMSE stems from a small set of samples with very large errors. Most predictions remain close to the ground truth as reflected by the low MAE, while these outliers inflate RMSE disproportionately.

This analysis supports the interpretation of the histogram shown in Figure 11, where a few bars far to the right (representing large errors) stand out against a background of primarily small-error samples. Such visualizations and sample analysis highlight the importance of not relying solely on RMSE as a summary metric and suggest that addressing these outlier cases—perhaps through more targeted data augmentation, better loss design, or model regularization—could significantly improve overall robustness.

The MAE assessment for existing solutions appears in Table 2. The MAE values for TransCrowd range from 66.1 to 105.1, while RMSE values range from 66.1 to 105.1 for Part A, followed by 9.3 to 16.1 for Part B. The MAE for CSRNet equals 68.2 and RMSE equals 115.0 on Part A and 10.6 to 16.0 on Part B. The MAE for SANet measures 67.0 and RMSE equals 104.5 for Part A and 8.4 and 13.6 for Part B. Our model produces a significantly lower MAE of 12.93 and RMSE of 204.52 on Part A, and 28.56 and 44.95 on Part B. The results demonstrate strong performance while showing the model's competitiveness across different density conditions.

The proposed MC-CNN with enhanced functionality performs effectively in essential areas by processing crowds at various density levels and extracting multi-scale spatial features. The MAE scores for SANet reach 67.0 for Part A and 8.4 for Part B. The model struggles to detect features within different crowd densities, particularly when numerous people gather in limited spaces. The MC-CNN achieves

Part A

Sample 1: GT = 1177.00, Pred = 1713.14, Error = 536.14



Sample 2: GT = 878.00, Pred = 440.23, Error = 437.77



Sample 3: GT = 393.00, Pred = 817.19, Error = 424.19



Sample 4: GT = 764.00, Pred = 345.09, Error = 418.91



Sample 5: GT = 554.00, Pred = 250.28, Error = 303.72

**Part B**

Sample 1: GT = 575.00, Pred = 300.79, Error = 274.21



Sample 2: GT = 521.00, Pred = 275.10, Error = 245.90



Sample 3: GT = 156.00, Pred = 355.92, Error = 199.92



Sample 4: GT = 340.00, Pred = 231.09, Error = 108.91



Sample 5: GT = 412.00, Pred = 306.54, Error = 105.46

**FIGURE 12.** Representative cases of prediction errors.

TABLE 2. Comparison table of MAE and RMSE on the test dataset.

Ref	Model	Part A		Part B	
		MAE	RMSE	MAE	RMSE
[32]	TransCrowd	66.1	105.1	9.3	16.1
[16]	SANet	67.0	20.3	104.5	13.6
[25]	CSRNet	68.2	115.0	10.6	16.0
	Our Model	12.93	204.52	28.56	44.95

this through its three parallel convolutional columns, which extract features at various spatial scales while demonstrating improved handling of diverse crowd density clusters.

The addition of a fusion layer to MC-CNN enables a better combination of multi-scale data into a unified crowd density map, which enhances generalization capabilities. This method ensures model accuracy with varying crowd densities. Real-life applications benefit from this approach because it improves performance when crowd conditions show significant changes. The MC-CNN demonstrates resistance to crowd pattern changes according to Li et al. [25], which leads to better performance in multiple challenging settings. Our neural network delivers exceptional MAE results (MAE 12.93 on Part A and MAE 28.56 on Part B) because it operates well in dense and sparse scenarios while extracting features at multiple scales.

In addition to the reported average Loss and MAE values, we provide 95% confidence intervals to assess the statistical reliability of the results. For the ShanghaiTech Part A dataset, the loss was 28.82 with a 95% confidence interval of [23.77, 33.88], and the MAE was 139.48 with a 95% confidence interval of [114.62, 164.34]. For Part B, the loss was 5.77 with a 95% confidence interval of [4.90, 6.64], and the MAE was 28.56 with a 95% confidence interval of [24.24, 32.88]. These intervals highlight the consistency and generalization capability of the proposed MC-CNN model across both sparse and dense crowd scenarios.

The training of the proposed crowd density estimation model across five epochs produced the results shown in Table 3 for Part A and Part B separately. The epoch number indicates the current training cycle position, which enables you to monitor training progress. The model demonstrates an excellent fit for the training data through Training Loss. Training MAE calculates the Mean Absolute Error, which occurs during training operations without considering the direction of errors. The loss calculation during validation helps determine the model's performance on unseen data.

From the table, we observe that:

1. The training loss decreases markedly from 9.76 in epoch 0 to 2.13 in epoch 4 (Part A), and from 7.26 to 0.38 in Part B, indicating effective model learning across both datasets. This finding aligns with the observations of Khan et al., who highlighted the significance of loss reduction in training effective crowd-counting models [2].

TABLE 3. Training MAE and Validation MAE across the epochs.

Part A				
Epoch	Training Loss (tr_loss)	Training MAE (mae)	Validation Loss (val_loss)	Validation MAE (val_mae)
0	9.76	47	15.11	73.11
1	2.99	14.75	9.11	44.02
2	2.55	12.55	2.92	14.23
3	2.07	10.11	2.22	10.7
4	2.13	10.4	3.54	17.46

Part B				
Epoch	Training Loss (tr_loss)	Training MAE (mae)	Validation Loss (val_loss)	Validation MAE (val_mae)
0	7.26	34.70	4.01	19.77
1	0.56	2.76	0.56	2.73
2	0.51	2.50	0.39	1.89
3	0.49	2.44	0.38	1.88
4	0.38	1.85	0.29	1.43

2. The training MAE significantly decreased from 47 in epoch 0 to 10.4 in epoch 4 (Part A), and from 34.70 to 1.85 in Part B, demonstrating enhanced accuracy in density predictions. This observation is consistent with the findings of Sam et al. and Boominathan et al., who documented comparable accuracy improvements via iterative training processes [4], [5].
3. The validation loss fluctuated, with an optimal performance recorded in epoch 3 for Part A (val_loss = 2.22) and epoch 4 for Part B (val_loss = 0.29), which suggests the model's strong generalisation capabilities. This is corroborated by the efficacy of deep learning techniques in managing variations in crowd density, as evidenced by the works of Cao et al. and Ryu and Song [16], [22].

Figure 13 shows the application of the MC-CNN model in Kazakhstan. The first set of images showcases a crowd in Khan-Shatyr mall in Astana with a corresponding predicted density map, the second set shows the same on Astana's square. The red colour denotes that these zones are overcrowded. Based on the predicted density maps, security officers could react accordingly whenever large gatherings happen, which could result in possible injuries or even deaths.

Overall, this section showed the major experimental results achieved throughout the study. The following section will discuss the potential implications of the findings, compare them with existing work, and discuss the strengths and limitations of our research.

V. DISCUSSION

The evaluation metrics for Part A and Part B demonstrate that the model learned effectively and generalized well to different crowd scenarios. The model reached a validation

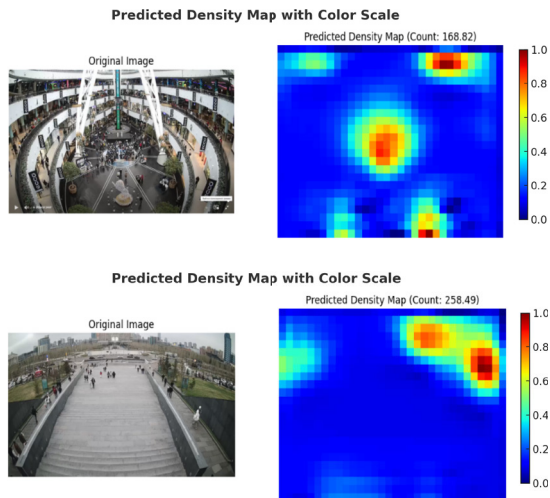


FIGURE 13. Application of the MC-CNN model in Kazakhstan.

MAE of 17.46 on Part A, which contains highly dense and complex crowd scenes, thus demonstrating its ability to handle extremely dense environments with severe occlusions and scale variations. The model achieved a validation MAE of 1.43 on Part B, which contains sparse to moderately dense crowds, thus proving its robustness in less challenging but highly variable real-world scenes. The proposed MC-CNN model demonstrates excellent generalization capabilities across various crowd densities by outperforming baseline models, including TransCrowd, SANet, and CSRNet. The consistent decrease in training and validation MAE throughout the epochs supports the model's learning efficiency and stability.

The MC-CNN architecture extends multi-column methods using a lightweight MobileNet-inspired backbone with three parallel convolutional columns and various kernel sizes. This design effectively captures multi-scale features and eases their fusion to comprehend spatial density variations in different crowd scenes. The proposed model is more accurate as it has a lower MAE than traditional CSRNet and SANet models, while striking a balance between prediction accuracy and model complexity.

The MC-CNN architecture has advantages in crowd density estimation. The model extracts features at different scales through parallel convolutional columns to identify varying densities' crowd patterns. The model's efficient design results in a relatively small model size and faster inference speed, making it suitable for real-time and edge deployments. The feature fusion system includes spatial data to raise crowd density prediction accuracy. The model demonstrates its strengths as per experimental results, and outperforms other advanced methods like TransCrowd, CSRNet, and SANet.

The system is accurate and suitable for real-time crowd estimation. The results show that the MC-CNN model processes images at around 45 milliseconds per frame when running on an NVIDIA A100 GPU. The model operates at approximately 22 FPS which proves its capability to work

close to real-time, aligns with most of the crowd estimation requirements.

Research studies [33] and [34] demonstrate that crowd movement patterns evolve slowly when large groups gather. The development of fast disordered crowd formations and subsequent dissolutions appears statistically improbable. The proposed system operates without needing complete frame rate processing of each video frame. The model can be used by real-world applications for crowd density monitoring at frame sampling rates between 1 and 5 FPS because these rates effectively identify significant crowd density alterations. The surveillance and crowd flow management analysis requires 200–500 milliseconds intervals, as shown in [35].

Video content analysis requires continuous processing, leading to substantial storage requirements, increased bandwidth usage, and computational needs according to [36] and [37]. The need for analyzing each video frame individually does not apply to crowd estimation because this task works effectively with reduced temporal resolution without affecting decision quality. Standard crowd monitoring needs cannot handle the extensive processing and storage needs of high-frame-rate video data because of its complex nature.

The model presents the optimal tradeoff between precision and speed. Real-time crowd detection of full-frame-rate video becomes unnecessary since crowd density changes occur gradually. Future research plans to enhance runtime performance through model pruning and quantization approaches, as well as dedicated AI hardware implementations. Future system improvements will expand the capabilities of urban surveillance, transport hubs, and emergency response systems under limited resource conditions.

The precise crowd counting function of the model allows effective proactive overcrowded areas' detection and population flow control, benefiting dangerous and crowded locations. This has implications for resource management, emergency response, and smart city development. The results show that implementing features at different scales enhances predictive performance. Future studies should evaluate various kernel size arrangements and architectural changes to improve performance and operational efficiency.

Empirical observations indicated that L2 regularization was not required during model training because the model successfully converged and generalized without such constraints. The weight norms and gradient plots of all epochs remained stable during the learning process, showing no signs of instability or overfitting. Crowd density estimation tasks achieve better results using architectural design elements (such as multi-column convolutional branches and adaptive receptive fields) instead of standard regularization approaches. Research should investigate how regularization methods impact robustness specifically when working with crowd data that contains imbalances or noise.

The MC-CNN method has several operational constraints. High-resolution input data requirements generate higher

computational needs that prevent real-time applications on low-power devices. This model achieves better accuracy through its multi-column design, yet its enhanced performance comes with increased complexity that hinders scalability in limited resource settings. The object separation capability of the model deteriorates in heavily obstructed scenarios because multiple-scale features fail to detect separate objects. The transfer learning process using MobileNet reduces computation time. It enhances inference speed, but weight dependencies may persist, which do not align perfectly with crowd data characteristics, thus requiring more fine-tuning or domain adaptation. Research must investigate lightweight models and attention mechanisms to enhance operational efficiency and robustness.

Our research plan involves validating the model using CrowdFaceYOLO [38] and other datasets to assess its performance in various high-density environments. The model requires testing against data it has not seen before. The model displays generalization in its initial tests, so we will conduct extra experiments to assess its performance under various crowd situations, including indoor and outdoor environments, and different lighting scenarios and occlusion levels. The model will achieve better results through domain adaptation techniques, which fine-tune with small amounts of labelled data to enhance transferability.

The design requires improvement and additional crowd data acquisition to establish reliable crowd predictions in various crowd situations. The study will evaluate real-time scenarios that require fast inference times and test the model with multiple datasets in real-world settings. Our team's high-accuracy localized estimation method should integrate with Hussain et al.'s [39] macroscopic urban analytics system to build a unified crowd monitoring system. Real-time data processing and fast inference times are needed for the system to analyze real-world data across multiple datasets in real-time.

Model interpretability is essential for deployment activities because safety-related decisions need clear transparency in critical domains. Grad-CAM, saliency maps, and Layer-wise Relevance Propagation (LRP) will be used to show where the model focuses its predictions in future research. Visual display of particular image areas affecting density prediction will improve trust and accountability.

The study introduces the modified MC-CNN method for accurate density prediction that can be generalised and executed in real time. The proposed model's efficiency, scalability, and interpretability improvement lay a solid foundation for future research and practical crowd monitoring.

The large prediction errors discovered in Part A dense scenes need either improved architecture tuning, better loss function approaches, or data augmentation methods to handle extreme cases more effectively. After dealing with the data outliers, the model becomes more robust with decreased RMSE discrepancy in dense crowd scenarios.

The validation loss and MAE show variability, especially in complex dense scenes of Part A, thus requiring additional research to enhance model robustness and reliability across different crowd scenarios. The model requires architectural design refinement, expanded training datasets, and improved real-time inference capabilities. The model involves testing with diverse real-world datasets and conditions for better generalization. The macroscopic urban analytics system by Hussain et al. [39] should be integrated with city-scale crowd management frameworks to establish a unified scalable crowd monitoring solution.

VI. CONCLUSION

The research introduces an improved MC-CNN architecture which combines MobileNet transfer learning with parallel convolutional branches for extracting multi-scale features. The model enhances training stability and actual applications via dynamic collation and a zero-padding strategy for handling inputs of varying sizes. The proposed model outperforms TransCrowd, CSRNet, and SANet as per MAE scores on the ShanghaiTech dataset. The experimental results show that the model can detect spatial crowd patterns in different environments. The research provides a scalable and accurate solution for crowd analysis, which overcomes previous methods' limitations and provides practical values for smart urban and public safety systems.

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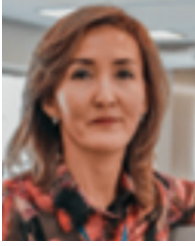
in SDU University. He also has industrial experience in Chocofamily Holding.



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